DAC Testnet

Official Enode Evolution Analysis — Infrastructure Rotation & Network Maturation

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Data Source	enodes.dachain.tech/testnet — official DAC enode registry
Report Type	Infrastructure Analysis — Network Topology Evolution & Operational Insight
Relates to	DAC Testnet (Inception) campaign — ongoing node operation field study

Executive Summary

This report documents and analyzes the evolution of the official DAC Testnet enode registry as observed since before the Inception campaign began and continuing through the Inception campaign. The official enode list at enodes.dachain.tech has undergone multiple changes since the campaign began, following the pattern:

2 → 5 → 9 → 10 → 6 enodes

Rather than indicating network instability, this pattern reflects a deliberate and healthy infrastructure lifecycle — from minimal bootstrap through active expansion and stress testing, toward a leaner, more stable operational baseline. The current published set of 6 enodes represents the latest operational snapshot, not a permanent final configuration, as further changes are expected as the testnet continues to evolve.

Observation	Enode Count	Operational Interpretation
Observation 1 — Initial Bootstrap	2	Minimal seed nodes for genesis and initial peer discovery
Obs. 2 — Core Stabilization	5	Core cluster expansion — redundancy and availability
Obs. 3 — Community Expansion	9	Multi-provider, multi-region stress testing
Obs. 4 — Highest Count So Far	10	Highest enode count observed so far in this analysis
Observation 5 — Consolidation (current)	6	Performance-based pruning — battle-tested nodes retained

Observation Methodology

This analysis is based on direct, ongoing monitoring of the official DAC enode registry at enodes.dachain.tech/testnet since before the Inception campaign began and continuing through the Inception campaign. Rather than deleting enodes that were removed from the official list, each new set was appended to the local node configuration (static-nodes.json and config.toml), allowing historical peer connectivity to be preserved while new enodes were integrated.

At the point of highest observed enode count (Obs. 4), the local peer list reached 14 active enodes — the 10 official enodes plus 4 retained from earlier observations — plus 1 internal peer from the dual-node WSL setup. This approach provided broader peer diversity than the official list alone and served as a natural historical record of enode rotation.

Observation-by-Observation Analysis

Observation 1 — Initial Bootstrap (2 Enodes)

Active nodes: 157.173.127.22 · 157.173.127.18

```
enode://37066e7d...f7fa8@157.173.127.22:28657
enode://623747311...b2f3@157.173.127.18:28657
```

The earliest documented state of the DAC Testnet bootstrap layer. Two nodes from the 157.173.127.x subnet — almost certainly core team infrastructure — served as the sole entry points for genesis distribution, initial peer discovery, and basic chain synchronization. Infrastructure is minimal and highly centralized by design at this stage.

- Core team seed nodes providing genesis entry point
- High centralization — intentional for early bootstrap control
- Both nodes from the same subnet, suggesting a single managed provider cluster

Observation 2 — Core Stabilization (5 Enodes)

Added: 157.173.127.31 · 157.173.127.30 · 157.173.127.21

```
enode://37066e7d...f7fa8@157.173.127.22:28657
enode://623747311...b2f3@157.173.127.18:28657
enode://a59112afa...25e3@157.173.127.31:28657
enode://9652549979...e8a1@157.173.127.30:28657
enode://27386ed9cc...098d@157.173.127.21:28657
```

Three additional nodes joined the 157.173.127.x cluster, expanding the bootstrap set to five nodes within the same provider infrastructure. This observation period focused on redundancy and availability — reducing single-node dependency while maintaining centralized control through a homogeneous subnet. All five nodes share the same IP block, indicating coordinated team infrastructure.

Notable: three of these nodes — 157.173.127.30, 157.173.127.21, and 157.173.127.31 — demonstrated exceptional longevity, persisting through multiple subsequent observation periods. This strongly suggests they form part of the core authority or validator backbone.

Observation 3 — Community & Multi-Region Expansion (9 Enodes)

Added: 156.67.104.212 · 144.91.74.181 · 206.189.127.204 · 161.97.89.27

A significant architectural shift: four nodes from distinct providers and geographic regions were introduced, breaking out of the single-subnet cluster model. This observation period indicates the team was actively stress-testing connectivity across diverse infrastructure providers and validating peer diversity.

Node IP	Likely Provider	Inferred Role
206.189.127.204	DigitalOcean	Bootnode / RPC support
161.97.89.27	Contabo (likely)	Support node
144.91.74.181	Contabo (likely)	Temporary infrastructure

156.67.104.212	European VPS	Regional relay node

Objectives: stress testing, geographic diversity validation, latency measurement, peer path redundancy.

Observation 4 — Highest Enode Count So Far (10 Enodes)

Added: 1.54.141.126 · 152.228.141.231 · 5.104.86.129 · 95.216.70.180

Highest enode count observed so far. Four additional nodes from distinct infrastructure categories were introduced, representing the broadest geographic and provider diversity observed throughout the campaign. This represents the highest enode count observed so far throughout this analysis.

Node IP	Likely Provider / Region	Inferred Role
1.54.141.126	Possibly Asian region (likely residential or dynamic IP)	External validator / community node test
152.228.141.231	OVHcloud — Europe	High-bandwidth EU relay bootnode
5.104.86.129	Less common VPS provider	Experimental support / community-operated
95.216.70.180	Possibly Hetzner — Germany / Finland	High-performance relay — temporary

The inclusion of 1.54.141.126 is particularly notable — its IP range characteristics suggest a residential or dynamic IP address, unlike the dedicated datacenter nodes in other observation periods. This may indicate the team was testing external or community-operated validator participation during this observation period.

Obs. 5 — Consolidation (6 Enodes, Current)

Retained: 157.173.127.30 · 157.173.127.21 · 206.189.127.204 · 152.228.141.231 · 5.104.86.129

New addition: 217.76.53.98

```
enode://52a3c25c...2272@217.76.53.98:28657
enode://9652549979...e8a1@157.173.127.30:28657
enode://4ff5ceea...ecf@152.228.141.231:28657
enode://2f4ed550...92f@5.104.86.129:28657
enode://27386ed9cc...098d@157.173.127.21:28657
enode://09b8b08d...5c4c@206.189.127.204:28657
```

Following the highest enode count observed so far, the official list was consolidated to 6 nodes. Eight nodes were removed and one new node (217.76.53.98) was introduced as a replacement. The retained nodes represent a curated mix of proven core infrastructure and selected multi-provider nodes that demonstrated stable performance throughout previous observations.

This is not a final configuration — it is the latest operational snapshot. Given the dynamic nature of an active testnet campaign, further changes remain probable.

Survivorship Analysis

Tracking which nodes persisted across phases reveals which infrastructure the DAC team considers most reliable and which served temporary roles.

Nodes That Survived to Obs. 5

Node IP	First Appeared	Observed In	Assessment
157.173.127.30	Obs. 2	Obs. 1 → 2 → 3 → 4 → 5	**Core backbone**
157.173.127.21	Obs. 2	Obs. 1 → 2 → 3 → 4 → 5	**Core backbone**
206.189.127.204	Obs. 3	3 → 4 → 5	**Proven relay node**
152.228.141.231	Obs. 4	4 → 5	**EU relay — selected**
5.104.86.129	Obs. 4	4 → 5	**Retained after evaluation**
217.76.53.98	Obs. 5	5 (new)	**Replacement / addition**

Possible Reasons for Nodes No Longer Published

Node IP	Last Observed	Possible Reason for Removal
157.173.127.22	Obs. 3	Replaced by stronger infrastructure within same cluster

157.173.127.18	Obs. 3	Replaced by stronger infrastructure within same cluster
157.173.127.31	Obs. 4	Core subnet consolidation — .30 and .21 retained
156.67.104.212	Obs. 3	Performance evaluation — did not meet retention criteria
144.91.74.181	Obs. 3	Temporary infrastructure — experimental purpose completed
161.97.89.27	Obs. 3	Temporary infrastructure — experimental purpose completed
95.216.70.180	Obs. 4	Hetzner relay — high-performance but not retained long-term
1.54.141.126	Obs. 4	Possible dynamic IP / residential — not suitable as permanent peer

Provider & Geographic Diversity Analysis

The current Obs. 5 enode set demonstrates a deliberate multi-cloud, multi-region distribution strategy — consistent with best practices for resilient blockchain infrastructure.

Node IP	Likely Provider	Region	Node Category
157.173.127.30	Core team infrastructure	—	**Authority / validator**
157.173.127.21	Core team infrastructure	—	**Authority / validator**
206.189.127.204	DigitalOcean	Global	**Bootnode / RPC relay**
152.228.141.231	OVHcloud	Europe	**EU relay node**
5.104.86.129	Independent VPS provider	—	**Support node**
217.76.53.98	To be confirmed	—	**New replacement node**

The multi-provider distribution significantly reduces single-point-of-failure risk. If one provider experiences downtime, the remaining nodes across different infrastructure providers maintain network accessibility. This is consistent with industry-standard practices for production blockchain bootstrap node management.

What This Timeline Reveals About DAC

The enode evolution pattern — 2 → 5 → 9 → 10 → 6 — is a textbook example of a well-managed blockchain testnet infrastructure lifecycle. Each transition reflects deliberate operational decisions rather than reactive responses to problems.

> **Active Network Management**

The DAC team is continuously monitoring, evaluating, and adjusting the bootstrap infrastructure. The repeated updates to the official enode list demonstrate hands-on involvement in network health.

Performance-Based Curation

Nodes are not removed arbitrarily. The retention of nodes like 157.173.127.30 and 157.173.127.21 across all five phases indicates a data-driven selection process based on uptime, sync consistency, and peer reliability.

Multi-Cloud Strategy Awareness

The deliberate inclusion of nodes across OVHcloud, DigitalOcean, Hetzner, and independent providers suggests an awareness of infrastructure diversity as a resilience consideration — though provider identities remain inferred from IP analysis and are not officially confirmed.

Iterative Infrastructure Optimization

The pattern of expand → test → consolidate is the correct approach for testnet development. Peak at 10 nodes allowed maximum coverage; consolidation to 6 reflects lessons learned from that period.

Testnet Is Actively Evolving

The introduction of a brand new node (217.76.53.98) in Obs. 5 — while others were removed — confirms the list is a living configuration, not a static reference. Further changes are expected and are a healthy sign of ongoing development.

Implications for Node Operators

Understanding the dynamic nature of the official enode list is important for anyone running a DAC Testnet node, particularly those participating in the Inception campaign.

Recommended Peer Configuration Strategy

Node	Recommended Static Peers
Windows Node	6 current official enodes + internal peer(s) if applicable
WSL Node	6 current official enodes + internal peer(s) if applicable

- Monitor enodes.dachain.tech/testnet periodically — the official list may change without announcement.
- Update static-nodes.json and config.toml whenever the official list changes.
- Old enodes that are no longer published can safely be retained — inactive nodes will simply be ignored by the client.
- Maintain internal peering between dual nodes (Windows ↔ WSL) for local redundancy independent of official peer availability.
- Do not treat the current 6-enode configuration as permanent — this is an operational snapshot during an actively evolving testnet.

Live Peer Topology — May 11, 2026

Beyond the official enode registry, active peer connectivity data was captured directly from both running nodes via admin.peers on May 11, 2026. This provides ground-truth evidence of which nodes are actually participating in the network — including nodes that are not listed on the official registry.

Node Status at Time of Observation

Node	eth.syncing	net.peerCount	Status
Windows Node (port 8545)	**false**	**13**	**Fully synced**
WSL Node (port 8546)	**false**	**13**	**Fully synced**

Verification commands used:

```
# Windows (PowerShell)
.\dacnode.exe attach ipc:\\.\pipe\gdacnode.ipc --exec "eth.syncing"
.\dacnode.exe attach ipc:\\.\pipe\gdacnode.ipc --exec "net.peerCount"
.\dacnode.exe attach ipc:\\.\pipe\gdacnode.ipc --exec "admin.peers"

# WSL / Linux
./dacnode attach ~/dac-linux/data/dac-chaindata-wsl/gdacnode.ipc --exec "eth.syncing"
./dacnode attach ~/dac-linux/data/dac-chaindata-wsl/gdacnode.ipc --exec "net.peerCount"
./dacnode attach ~/dac-linux/data/dac-chaindata-wsl/gdacnode.ipc --exec "admin.peers"
```

Active Peers — Identity Mapping

The admin.peers output revealed the self-declared identities of each connected node. This provides significantly more context than IP addresses alone, confirming the roles and classifications of nodes across the network — including official authority nodes, community VPS operators, and previously undocumented internal nodes.

Node Identity (name)	IP Address	Official List	Classification
DAC Testnet Authority 1	157.173.127.31	Obs. 2–4 (removed Obs. 5)	**Authority node**
DAC Testnet Authority 2	157.173.127.30	Obs. 2–5 (current)	**Authority node**
DAC Testnet Authority 3	157.173.127.21	Obs. 2–5 (current)	**Authority node**
DAC Testnet RPC 03	84.46.253.182	Never published	**Internal RPC node**
DAC-Node 05	206.189.127.204	Obs. 3–5 (current)	**Official relay**
gdacnode (legacy build)	152.228.141.231	Obs. 4–5 (current)	**Official relay**
whale-vps1	217.76.53.98	Obs. 5 (current)	**Community VPS**
whale-vps2	156.67.104.212	Obs. 3–4 (removed)	**Community VPS**
whale-vps3	161.97.89.27	Obs. 3 (removed)	**Community VPS**
x0rabbit	5.104.86.129	Obs. 4–5 (current)	**Community node**
SAPInode	213.136.82.243	Never published	**Unlisted active**

Fertal	95.216.70.180	Obs. 4 (removed)	**Community VPS**

Key Discoveries from Peer Identity Data

1. Authority Node Naming Convention Confirmed

The admin.peers output confirmed the naming of three nodes that form the backbone of the DAC Testnet authority layer: DAC Testnet Authority 1, 2, and 3. These are the nodes on the 157.173.127.x subnet that have been the most consistent presence throughout all observed phases. Their naming convention directly confirms they serve as consensus authority nodes rather than general bootnodes.

2. DAC Testnet RPC 03 — Unlisted Internal Node

A node identified as "DAC Testnet RPC 03" (84.46.253.182) is actively peering with both nodes but has never appeared on the official enode registry at enodes.dachain.tech. This strongly suggests the DAC team maintains internal infrastructure nodes that participate in the network but are intentionally not published as public bootstrap peers. Its "RPC 03" naming implies the existence of a broader internal RPC node infrastructure beyond what is publicly documented.

3. SAPInode (213.136.82.243) — Unlisted Active Peer

A node named "SAPInode" at 213.136.82.243 appears in the active peer list of both nodes but was never included in the official enode list at any point. This may represent a community-operated or partner node that independently connected to the network via peer discovery, rather than through the official static bootstrap list.

4. Removed Nodes Still Actively Peering

Several nodes that were removed from the official registry continue to actively participate in the network: DAC Testnet Authority 1 (157.173.127.31), whale-vps2 (156.67.104.212), Fertal (95.216.70.180), and whale-vps3 (161.97.89.27). This confirms that removal from the published list does not mean a node is shut down — it may continue running and peering via cached discovery or static configuration on other nodes.

5. Legacy Build at 152.228.141.231

The node at 152.228.141.231 (identified simply as "gdacnode" with no custom identity) is running a different build version: v1.10.5-stable-29a62a5e with Go 1.21.13, compared to the v1.10.5-stable-e4023222 / Go 1.25.1 used by most other active nodes. This suggests it has not been updated to the latest binary, which may explain why it carries no custom identity name — it was likely deployed without the identity flag or before the naming convention was established.

Real-Time Enode Update: 6 → 8 Enodes

Between May 10 and May 11, 2026 — within a single day, and precisely while this report was being prepared — the official enode count at enodes.dachain.tech changed from 6 to 8. This real-time update directly validates the primary assertion of this report: the official enode list is a living configuration, not a static endpoint.

Date	Official Count	Status
May 10, 2026	**6 enodes**	Obs. 5 consolidation — baseline observed
May 11, 2026	**8 enodes**	Expansion within 24 hours — hypothesis confirmed

This single-day delta from 6 to 8 nodes within an active testnet campaign confirms that the DAC team continues to make rapid infrastructure adjustments. Node operators should not treat any snapshot of the official list as permanent, and periodic monitoring of enodes.dachain.tech remains essential for maintaining optimal peer connectivity.

Observer Node Configuration — May 11, 2026

The following configuration reflects the current dual-node setup used to conduct this observation. Notable upgrades from earlier reports include increased maxpeers capacity, integrated metrics and profiling endpoints, and explicit config.toml loading.

Windows Node

```
.\dacnode.exe
  --identity "JERUZZALEM WINDOWS"
  --config "D:\...\chaindata\gdacnode\config.toml"
  --testnet --syncmode fast
  --miner.etherbase 0x2a01EAA3175a2796acD8F64fCe6dD6488D39D3Af
  --datadir "D:\...\chaindata"
  --port 28657 --maxpeers 25
  --nat extip:192.168.100.7
  --http --http.addr 0.0.0.0 --http.port 8545
  --http.api eth,net,web3,txpool,personal
  --metrics --metrics.addr 0.0.0.0 --metrics.port 6060
  --pprof --pprof.addr 0.0.0.0 --pprof.port 7070
  --verbosity 3
```

WSL / Linux Node

```
./dacnode
  --identity "JERUZZALEM LINUX"
  --config ~/dac-linux/data/dac-chaindata-wsl/gdacnode/config.toml
  --testnet --syncmode fast
  --miner.etherbase 0x870ad63acc507cdfd878F170606d19ae78988AFE
  --datadir ~/dac-linux/data/dac-chaindata-wsl
  --port 30304 --maxpeers 25
  --nat extip:192.168.100.7
  --http --http.addr 0.0.0.0 --http.port 8546
  --http.api eth,net,web3,txpool,personal
  --http.vhosts "*" --http.corsdomain "*"
  --metrics --metrics.addr 0.0.0.0 --metrics.port 6061
  --pprof --pprof.addr 0.0.0.0 --pprof.port 7071
  --cache 4096 --verbosity 3
```

Note on Dual Wallet Usage

This setup utilizes two separate wallet addresses — one per node. This is not a multi-account strategy for reward maximization, but a direct consequence of running two independent nodes, each requiring its own etherbase address for operational correctness. One wallet was created specifically following the wallet creation guidance published in the DAC official documentation (docs.dachain.tech), while the other is a personal testnet wallet used across various blockchain projects — particularly during testnet participation phases of different networks. Throughout this technical report series, the dual-wallet setup has been essential for conducting cross-wallet transaction testing, RPC validation, and interoperability analysis — all of which are well-documented in the accompanying technical reports.

Future Outlook — Possible Evolution Scenarios

The 6 → 8 change observed within a single day (May 10–11) has already confirmed the most likely scenario. The following outlook is now updated with this real-time data:

Scenario	Description	Status
Expansion to 8 nodes	Was projected as high probability	**Confirmed — May 11**
Further expansion (10–12 nodes)	Additional nodes added as network load increases	**Probable**
IP rotation within same count	Count stable, individual IPs replaced	**Ongoing**
Consolidation	Return to fewer, higher-quality nodes	**Possible later**

Conclusion

The evolution of the DAC Testnet official enode registry from 2 to a currently published set of 8 nodes — through observed states of 5, 9, and 10 — is a clear and well-structured infrastructure lifecycle. Far from being a sign of instability, this pattern demonstrates that the DAC team is actively and deliberately managing their network topology.

The retained nodes — particularly the long-surviving 157.173.127.30 and 157.173.127.21 — represent battle-tested core infrastructure. The multi-provider composition of the current set (spanning DigitalOcean, OVHcloud, and independent providers) reflects a mature approach to redundancy and geographic distribution.

Most importantly, the current 6-enode configuration should be understood as the latest operational snapshot in an ongoing process — not a final configuration. The DAC network is actively evolving, and the enode list will continue to reflect that evolution. Node operators are advised to treat the official registry as a living reference and monitor it regularly.

Fewer published nodes does not mean a weaker network.

It means the team has identified and retained only those nodes that meet the required quality criteria — including sync consistency, peering reliability, P2P stability, and uptime.

Community Resources — Dual Node Setup & Tooling

The dual-node single-machine setup used throughout this observation series has been fully documented and open-sourced for the DAC community. The following resources may be useful for node operators facing similar infrastructure challenges or looking to extend their monitoring and observability stack.

CGNAT Dual Node Setup

For node operators running under CGNAT network constraints or interested in replicating a dual-node (Windows + WSL) architecture on a single machine, the full setup — including startup scripts, peer configuration, topology documentation, and this report series — is available at:

```
https://github.com/EdLWEISS186/dac-dual-node-cgnat-setup
```

DAC Node Dashboard

A Windows desktop application for running, monitoring, and logging both nodes simultaneously from a single interface. Features include dual node control, graceful shutdown, real-time log streaming with color coding, live sync and peer monitoring, and structured logging to file. Available at:

```
https://github.com/EdLWEISS186/dac-dual-node-cgnat-setup/tree/main/DAC-Node-Dashboard
```

Infrastructure Monitoring Stack

A locally-hosted monitoring stack built with Prometheus, Grafana, and Blackbox Exporter for observing real-time operational status of both nodes. Configuration files and deployment instructions are available at:

```
https://github.com/EdLWEISS186/dac-dual-node-cgnat-setup/tree/main/monitoring
```

Technical Reports Archive

All technical reports from the Inception campaign — including the four preceding reports covering RPC availability, infrastructure validation, EIP-1559 compatibility, and NFT frontend desynchronization — are archived at:

```
https://github.com/EdLWEISS186/dac-dual-node-cgnat-
setup/tree/main/Testnet_(Inception)_Technical_Reports
```